

Sivabalan D

AI Applications Engineer | Full-Stack Developer

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Summary

Aspiring AI Engineer and Full-stack Developer pursuing a B.Tech in Information Technology (2027), with two web development internships. Builds tested, containerized applications with Python, FastAPI, React, and PostgreSQL, and applies PyTorch, scikit-learn, and LangChain across anomaly detection, LLM assistants, and Zero Trust security.

Education

Manakula Vinayagar Institute of Technology, Puducherry, India 2023 – 2027 (Expected)
Bachelor of Technology (B.Tech) in Information Technology CGPA: 7.4

SRVS National Higher Secondary School, Karaikal HSC: 64% (2023) | SSLC: Pass (2020)

Skills

Programming Languages: Python, Java, SQL

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Vite, Responsive Web Design

Backend & APIs: FastAPI, REST APIs, JWT, OAuth

AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain, OpenCV, DeepFace, Deep Learning, NLP, LLMs, Anomaly Detection

Databases & DevOps: PostgreSQL, MySQL, Docker, Pytest, Git, GitHub, Vercel, VS Code

CS Fundamentals: Object-Oriented Programming (OOP), DBMS

Experience

No-Code Web Development Intern – Gari Tech, Karaikal, Puducherry Jun 2025 – Jul 2025

- Redesigned a client's WordPress site with Elementor on XAMPP, focusing on responsive UX.
- Built an interactive online booking system to streamline reservations and scheduling.
- Applied on-page SEO best practices to improve site structure and search discoverability.

Web Development Intern – Ruddo, Karaikal, Puducherry Nov 2024 – Jan 2025

- Developed responsive, UI/UX-focused web interfaces using HTML5, CSS3, JavaScript, and Bootstrap.
- Built a full-stack Online Doctor Consultation web application.
- Ensured cross-browser compatibility and optimized front-end layouts and styling.

Projects

Trust Shield AI: Explainable Zero Trust Access github.com/sivabalan-05/ztna-adaptive-trust

- Built a Zero Trust Network Access (ZTNA) platform that turns 6 weighted signals (identity, device, network, behavior, location, time) into a 0–100 trust score driving allow, step-up MFA, or block decisions.
- Detected behavioral anomalies with an Isolation Forest model and re-scored every active session every 30 seconds, enforcing TOTP MFA, JWT authentication, and role-based access control (RBAC).
- Built an explainable score breakdown and a SHA-256 hash-chained audit log; validated with 345 automated tests and 7 attack scenarios.
- Tech Stack:** Python, FastAPI, scikit-learn, React, TypeScript, Tailwind CSS, PostgreSQL, Redis, Docker, Pytest

WeatherGPT – AI Weather Assistant github.com/sivabalan-05/weatherGPT

- Built a weather intelligence platform with real-time Open-Meteo data, 24-hour and 7-day forecasts, an interactive Leaflet map, and climate trend charts.
- Developed a natural-language assistant that extracts location, time, and intent from queries and uses Gemini or OpenAI to explain verified weather data without hallucinating values.
- Implemented IMD-aligned alerts for 6 severe weather hazards, advisories for 7 crops, voice input, and English, Hindi, Telugu, and Tamil support.
- Tech Stack:** Python, FastAPI, React, TypeScript, Tailwind CSS, Leaflet, Recharts, Gemini API, OpenAI API

Languages

English (Professional) | Tamil (Native) | Hindi (Intermediate)